

A Weibull Counterexample to a Proposed Tail-Ratio Inequality

Working note

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Abstract

Problem 26 in Stefan Steinerberger's *Some Open Problems* asks whether the exponential distribution is extremal for a ratio comparing “one large jump” with “two moderate jumps.” We give an elementary counterexample to the mean-normalized version. If X, Y are independent Weibull random variables with survival function $\mathbb{P}(X \geq x) = \exp(-x^{3/2})$, then for every $z > 0$ the ratio in question is strictly smaller than $2z$, whereas $2z/\mathbb{E}X > 2z$. The proof uses only a quadratic convexity estimate and Hölder's inequality. The same example also shows that the natural growth scale of the ratio need not be linear: it is asymptotic to $\sqrt{3\pi} z^{3/4}$.

Idea of the proof. The exponential case is linear because memorylessness makes every decomposition of $2z$ contribute equally. For the Weibull tail $e^{-x^{3/2}}$, strict convexity penalizes unequal decompositions quadratically, so the integral concentrates in a Gaussian window around $x = z$, producing $z^{3/4}$, not linear, growth.

1 The question

For independent, identically distributed nonnegative random variables X, Y , write

$$R_X(z) = \frac{\mathbb{P}(X + Y \geq 2z, \min(X, Y) \leq z)}{\mathbb{P}(X + Y \geq 2z, \min(X, Y) \geq z)}.$$

Motivated by the identity $R_X(z) = 2z/\mathbb{E}X$ for an exponential law, Steinerberger asked whether every absolutely continuous, non-compactly supported law on $[0, \infty)$ with finite mean admits some $z > 0$ such that

$$R_X(z) \geq \frac{2z}{\mathbb{E}X}. \quad (\text{Q})$$

2 Counterexample

Theorem 1. *Let X, Y be independent with*

$$\mathbb{P}(X \geq x) = e^{-x^{3/2}}, \quad f(x) = \frac{3}{2}\sqrt{x} e^{-x^{3/2}} \quad (x \geq 0).$$

Then, for every $z > 0$,

$$\frac{R_X(z)}{z} \leq 3 \left(\frac{4\pi}{45} \right)^{1/3} < 2 < \frac{2}{\mathbb{E}X}.$$

Consequently, no $z > 0$ satisfies (Q).

Proof. Because the law is continuous, the choice of weak or strict inequalities on the boundary is immaterial. The denominator is $\mathbb{P}(X \geq z, Y \geq z) = e^{-2z^{3/2}}$. By symmetry, the numerator is $2 \int_0^z f(x) \mathbb{P}(Y \geq 2z - x) dx$.

Thus, after setting $x = zt$,

$$R_X(z) = 3z^{3/2} \int_0^1 \sqrt{t} \exp[-z^{3/2}\psi(t)] dt, \quad \psi(t) = t^{3/2} + (2-t)^{3/2} - 2. \quad (1)$$

Put $s = 1 - t$. We claim

$$(1-s)^{3/2} + (1+s)^{3/2} - 2 \geq \frac{3}{4}s^2, \quad 0 \leq s \leq 1. \quad (2)$$

Indeed, the second derivative of the left side is $\frac{3}{4}((1-s)^{-1/2} + (1+s)^{-1/2}) \geq \frac{3}{2}$, by convexity of $u \mapsto u^{-1/2}$, and both sides and their first derivatives agree at $s = 0$.

Set $a = \frac{3}{4}z^{3/2}$. Equations (1)–(2) give

$$\frac{R_X(z)}{z} \leq 3 \left(\frac{4a}{3} \right)^{1/3} \int_0^1 (1-s)^{1/2} e^{-as^2} ds. \quad (3)$$

Hölder's inequality with conjugate exponents 3 and 3/2 yields

$$\int_0^1 (1-s)^{1/2} e^{-as^2} ds \leq \left(\int_0^1 (1-s)^{3/2} ds \right)^{1/3} \left(\int_0^1 e^{-(3/2)as^2} ds \right)^{2/3} \leq \left(\frac{\pi}{15a} \right)^{1/3}. \quad (4)$$

Combining (3) and (4), $R_X(z)/z \leq 3(4\pi/45)^{1/3} < 2$; the final strict inequality is equivalent to $\pi < 10/3$. Finally,

$$\mathbb{E}X = \int_0^\infty e^{-x^{3/2}} dx = \Gamma\left(\frac{5}{3}\right) < 1,$$

where the last inequality follows from strict log-convexity of Γ between $\Gamma(1) = \Gamma(2) = 1$. Hence $2/\mathbb{E}X > 2$. $\square$

3 The growth rate is sublinear

As $s \rightarrow 0$,

$$(1-s)^{3/2} + (1+s)^{3/2} - 2 = \frac{3}{4}s^2 + O(s^4).$$

Applying the standard endpoint Laplace estimate to (1) gives

$$\int_0^1 \sqrt{t} e^{-z^{3/2}\psi(t)} dt \sim \sqrt{\frac{\pi}{3}} z^{-3/4},$$

and therefore

$$R_X(z) \sim \sqrt{3\pi} z^{3/4} \quad (z \rightarrow \infty). \quad (5)$$

Thus the Feldheim–Feldheim divergence $\limsup R_X(z) = \infty$ is fully compatible with every positive linear lower-rate conjecture failing.

Remark 1 (Scope). *This settles only the mean-normalized question. It does not settle the separate median-normalized inequality in Problem 26.*

Novelty status

This result was derived on 20 July 2026 after computational reconnaissance, but the proof above is entirely analytic. A targeted search located the open question in Steinerberger's 2018 paper and in the October 2023 problem list, and found no later paper resolving it. This is a preliminary novelty check, not independent confirmation; the claim should be checked by the problem's author or a probability specialist before being advertised as new.

References

- [1] N. D. Feldheim and O. N. Feldheim, *Mean and Minimum of Independent Random Variables*, arXiv:1609.03004, revised 2020.
- [2] S. Steinerberger, *A Sharp Estimate for Probability Distributions*, Statistics & Probability Letters 155 (2019), 108584; arXiv:1801.00663.
- [3] S. Steinerberger, *Some Open Problems*, Problem 26, version dated 31 October 2023.